

THE COLLATZ CONJECTURE

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[Github](#)

1. Introduction

1.1 Background

The Collatz Conjecture, which is also referred to as the $3n+1$ conjecture has been a standing puzzle, in number theory ever since Lothar Collatz introduced it in 1937. Despite its nature the conjecture remains unsolved posing a unique challenge to the mathematical community.

1.2 Problem Statement

According to this conjecture by starting with any integer we can generate a sequence using iterative rules until we eventually reach the value of 1. Although these rules may seem simple they lead to unpredictable behavior, in practice.

$$a_n = \begin{cases} \frac{1}{2} a_{n-1} & \text{if } a_n \text{ is even} \\ 3a_{n-1} + 1 & \text{if } a_n \text{ is odd} \end{cases}$$

2. Historical Context

2.1 Origins of the Collatz Conjecture

Lothar Collatz introduced the eponymous conjecture in 1937 during a series of mathematical discussions. The conjecture's accessibility, coupled with its deceptively simple rules, garnered initial interest from mathematicians and non-mathematicians alike.

2.2 Early Investigations

The conjecture's first wave of exploration occurred in the 1950s and 1960s, marked by manual calculations and computer simulations to observe the behavior of sequences generated by the conjecture. Despite these early efforts, a rigorous proof or disproof remained elusive.

2.3 Proliferation of Interest

As researchers delved deeper into the Collatz Conjecture, the mathematical community witnessed a proliferation of interest in the latter half of the 20th century. The conjecture's innate simplicity, juxtaposed with its resistance to formal analysis, led to an increasing number of investigations.

2.4 Computational Efforts

With the advent of computers, the Collatz Conjecture became the subject of extensive computational studies. Mathematicians leveraged computational tools to test the conjecture for a vast range of initial values, revealing intricate and seemingly chaotic behaviors within the sequences.

2.5 Status as an Open Problem

The Collatz Conjecture maintains its status as an open problem. Despite decades of exploration, no general proof or disproof has been established. The apparent simplicity of its formulation continues to intrigue mathematicians, making it a persistent and captivating challenge in the realm of number theory.

3. Addressing The Gaps in Existing Solutions :

In the pursuit of resolving the Collatz Conjecture, various attempts have been made to explore and understand the patterns exhibited by its sequences. Existing solutions often involve extensive computational analyses, leveraging advanced algorithms and computing resources. While these efforts have provided valuable insights into the behavior of the Collatz sequence for specific initial values, gaps persist in achieving a comprehensive and universally applicable proof. The conjecture's resistance to conventional mathematical techniques remains a formidable challenge, and the intricacies of even seemingly straightforward cases contribute to the elusive nature of a conclusive solution. Additionally, the reliance on computational methods, while informative, introduces the possibility of undetected patterns or anomalies that may be overlooked in exhaustive analyses. Therefore, addressing these gaps necessitates a nuanced approach that integrates theoretical rigor with computational efficiency, emphasizing the need for innovative methodologies to unravel the mysteries embedded in the Collatz Conjecture.

4. Goal and Importance of Solving the Conjecture

Our primary goal is to establish a universal proof that, for all integers n in the set of integers ($\mathbb{Z}$), the Collatz sequence will inevitably converge to the value 1. This pursuit reflects a broader exploration into the nature of mathematical patterns.

5. Comprehensive Methodology

5.1 Even Numbers Convergence in Collatz :

Idea :

In the proposed matrix of even numbers, a systematic approach is established to methodically examine the convergence patterns inherent in the Collatz Conjecture for even integers. The matrix is meticulously structured to showcase a verifiable relationship, wherein an even number, denoted as n is expressed as the product of a previously proven small odd number m and 2 raised to the power of a natural number i . Mathematically, this relationship is represented as $n = m \cdot 2^i$. The significance of this formulation lies in the transitive property of convergence. If an even number n can be precisely expressed in the specified form, it unequivocally returns to the proven small odd number m , thereby establishing the provability of n within the Collatz Conjecture. This approach provides a rigorous and systematic means of verifying convergence patterns for even numbers, enhancing our understanding of their behavior in the Collatz sequence.

Mathematical Formula Description:

The mathematical formula encapsulating the even matrix is elegantly expressed as

$$n = m \cdot 2^i$$

where:

n : denotes the even number under scrutiny in the Collatz Conjecture.

m: represents a previously proven small odd number to which *n* converges.

i : is a natural number, ranging from 0 onwards, serving as the positional indicator of *n* in the line of convergence to *m*.

In essence, this formula embodies the systematic relationship between even numbers and their convergence patterns. The parameter *i* delineates the sequential steps within the Collatz sequence, articulating the position of *n* concerning its return to the proven odd number *m*. This succinct formula encapsulates the fundamental structure of the even matrix, providing a clear and concise representation of the Collatz Conjecture's convergence dynamics for even integers.

3	6	12	24
5	10	20	40
7	14	28	36
9	18	36	72
11	22	44	88

Each row of the matrix corresponds to the sequential application of the Collatz sequence for an odd number, denoted as *m* , and its successive even multiples, (n_1, n_2, n_3) . For instance, considering the first row where $m = 3$,we observe that $n_i = m \cdot 2^i$.

Illustrative Explanation:

m	$n_1 = m \cdot 2^1$	$n_2 = m \cdot 2^2$	$n_3 = m \cdot 2^3$
3	6	12	24

Collatz sequence for (3): (10, 5, 16, 8, 4, 2, 1)

Collatz sequence for (6): (3, **10**, **5**, **16**, **8**, **4**, **2**, 1)

Collatz sequence for (12): (6, 3, **10**, **5**, **16**, **8**, **4**, **2**, 1)

Collatz sequence for (24): (12, 6, 3, **10**, **5**, **16**, **8**, **4**, **2**, 1)

Notably, all these sequences converge to the same repeating sequence: (3, 10, 5, 16, 8, 4, 2, 1). This common sequence underscores a pattern wherein all numbers in this row can be expressed as $3 \cdot 2^i$ and eventually return to (3). This repetition highlights the systematic nature of the Collatz sequence for even numbers in the matrix.

Limitation of the Even Numbers Matrix:

The even numbers matrix, while providing a systematic approach to verify convergence patterns, encounters a limitation when confronted with even numbers that return to large odd numbers not yet proven to converge to the fundamental value of 1 in the Collatz sequence. Specifically, the challenge arises when attempting to validate the convergence of even numbers to large odd numbers that lack a confirmed path to the ultimate convergence point. This limitation introduces a degree of uncertainty in the matrix's effectiveness, particularly in cases where the collatz sequence does not exhibit a clear and proven path to the established small odd numbers. Consequently, the even numbers matrix may face difficulties in comprehensively addressing the entirety of the Collatz Conjecture landscape, particularly in scenarios involving larger odd numbers with unverified convergence patterns.

5.2 Odd Numbers Convergence in Collatz :

In response to the limitation encountered by the even numbers matrix, the Matrix of Odd Numbers, denoted as , is introduced as a strategic solution. This matrix aims to establish a crucial link between large odd numbers, lacking a confirmed convergence path, and smaller odd numbers that have been empirically proven to converge within the Collatz sequence.

Idea:

The Odd Numbers Matrix encompasses odd numbers that cannot be expressed in the form $(3(2n + 1))$. These odd numbers, including (5, 7, 11, 13, 17...), serve as anchor points within the Collatz Conjecture. They act as pivotal nodes for the convergence of larger odd numbers that face uncertainty in their convergence patterns. Through a carefully designed iterative process, these large odd numbers are guided back to smaller odd numbers already established as convergent, ensuring a coherent and verified pathway back to the fundamental convergence point of 1. The introduction of the Odd Numbers Matrix thus offers a nuanced and strategic approach to overcoming the challenges posed by larger odd numbers within the Collatz sequence.

Mathematical Formula and Abstraction from the Even Matrix :

In the context of the Odd Numbers Matrix, a pivotal mathematical property is harnessed to facilitate convergence. For any odd number m within the Collatz sequence, a specific operation is applied: $3m + 1$. This operation is chosen strategically as it guarantees the production of an odd number after the first step and an even number after the second. Leveraging this property, the obtained even number can be seamlessly reintegrated into the Even Numbers Matrix, which has been established as a repository for proven convergent even numbers.

This mathematical property serves as a bridge between the odd and even matrices, enabling the redirection of larger odd numbers, potentially lacking a clear convergence path, back to the proven convergence patterns encapsulated in the Even Numbers Matrix. By abstracting this principle from the even matrix, the Odd Numbers Matrix introduces a dynamic mechanism for guiding larger odd numbers back to smaller odd numbers that have already been empirically proven to converge within the Collatz sequence. This strategic abstraction reinforces the systematic nature of the proposed approach, enriching the understanding and applicability of the Odd Numbers Matrix.

Illustrative Explanation:

The provided Odd Numbers Matrix, exemplifying the proposed methodology, is structured as follows:

1	2	4	8	16
$(4-1)/3=1$			$(16-1)/3=5$	
3	6	12	24	48
5	10	20	40	160
$(10-1)/3=3$			$(40-1)/3=13$	
7	14	28	56	112
$(28-1)/3=9$			$(112-1)/3=37$	
9	18	36	72	144

Mathematical Formula Description:

5	3	13	53
7	9	37	149
11	7	29	117
13	17	69	277

This column comprises odd numbers excluding those expressible as $3(2n + 1)$. Notably, odd numbers in this column do not have direct paths returning to them within

For each odd number in the first column denoted as m , the second column represents $\frac{(m \cdot 2^i) - 1}{3}$, where $i = \begin{cases} 1 & \text{if } (m-1) \text{ not divisible by } 3 \\ 2 & \text{if } (m-1) \text{ divisible by } 3 \end{cases}$. This formulation ensures a direct return to smaller odd numbers.

Each entry is calculated as $\frac{(m \times 2^i) - 1}{3} + (m \times 2^i)$ where m is the return odd number from the first column, and i is determined as before (with step $i = i + 2$). This iterative process captures the successive steps of the Collatz sequence, ensuring a systematic return to smaller odd numbers.

Optimization of the Odd Numbers Matrix:

To optimize the Odd Numbers Matrix, a discerning pattern is identified: the presence of numbers that return to a smaller odd number. In particular, the matrix exhibits that the number 5 returns to 1. Leveraging this insight, an optimization strategy is applied by consolidating the lines of numbers that return to the same smaller odd number.

For instance, considering the lines corresponding to 1 and 5, where the latter returns to the former, the optimization involves adding the entries of the line for 5 to the corresponding entries of the line for 1. Subsequently, the line for 5 is removed, streamlining the matrix while preserving the convergence information.

1	5	21	85
5	3	13	53
7	9	37	149

This optimization process is extended to all instances where a larger odd number returns to the same smaller odd number. By systematically consolidating these lines, the optimized Odd Numbers Matrix retains the essential convergence patterns while enhancing efficiency and reducing redundancy. This streamlined representation not only simplifies the matrix but also facilitates a more concise and informative depiction of the Collatz sequence dynamics for odd numbers.

Optimizing Odd Numbers Matrix through Tree Abstraction:

The optimization strategy involves the creation of a tree structure where all numbers (children nodes) converge to the root, which is the number 1. The key premise lies in proving the convergence of numbers by demonstrating that their Collatz sequences lead to the root. Specifically, the focus is on the first set of children, denoted as m with the Collatz operation $3m + 1 = 2^i$, where i is an even number.

The logic unfolds as follows:

1. Collatz Operation on Children of 1:

- For the first set of children m of 1, applying the Collatz operation $3m + 1$ results in 2^i , where i is an even number.

2. Iterative Collatz Transformation:

- Upon applying the Collatz sequence to these even numbers 2^i , the expression 2^{i-j} is obtained, where $i - j = 0$, ultimately yielding $2^{i-j} = 1$.

3. Conclusive Provenance:

- The consequential outcome signifies that all numbers abstracted in the tree from the initial children of 1 are proven, as their iterative Collatz transformations consistently lead to the fundamental value of 1.

This optimization technique streamlines the proof process by abstracting numbers in the tree, demonstrating their provability through a logical sequence of Collatz operations that invariably converge to the root. As a result, the entire subtree stemming from the children of 1 is conclusively established as proven within the Collatz Conjecture.

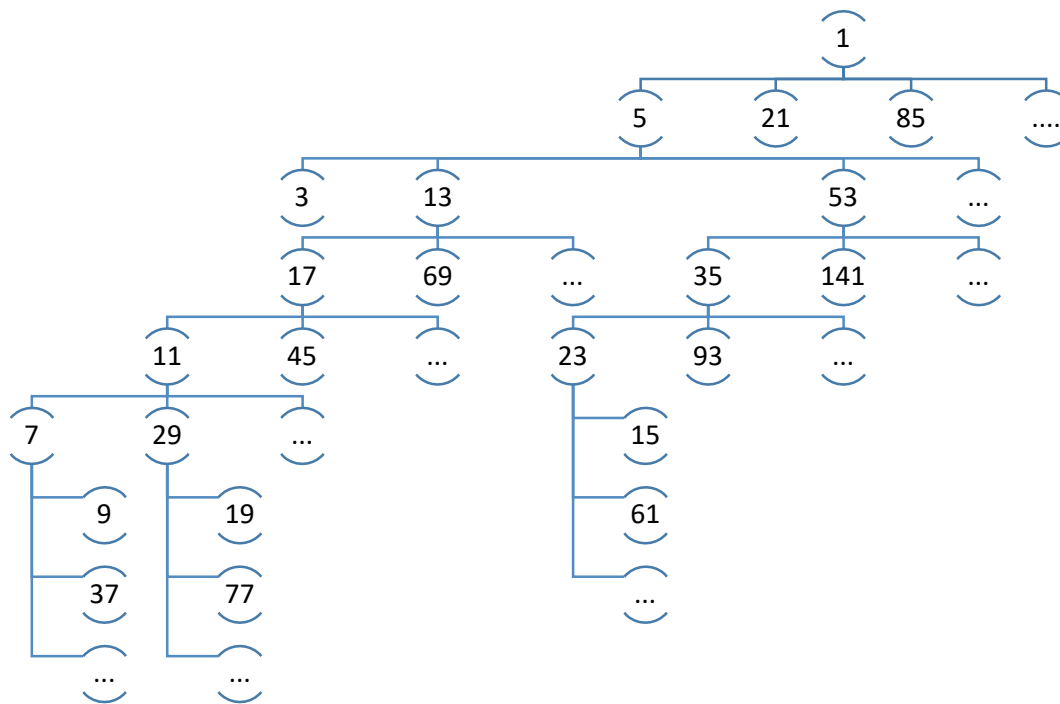
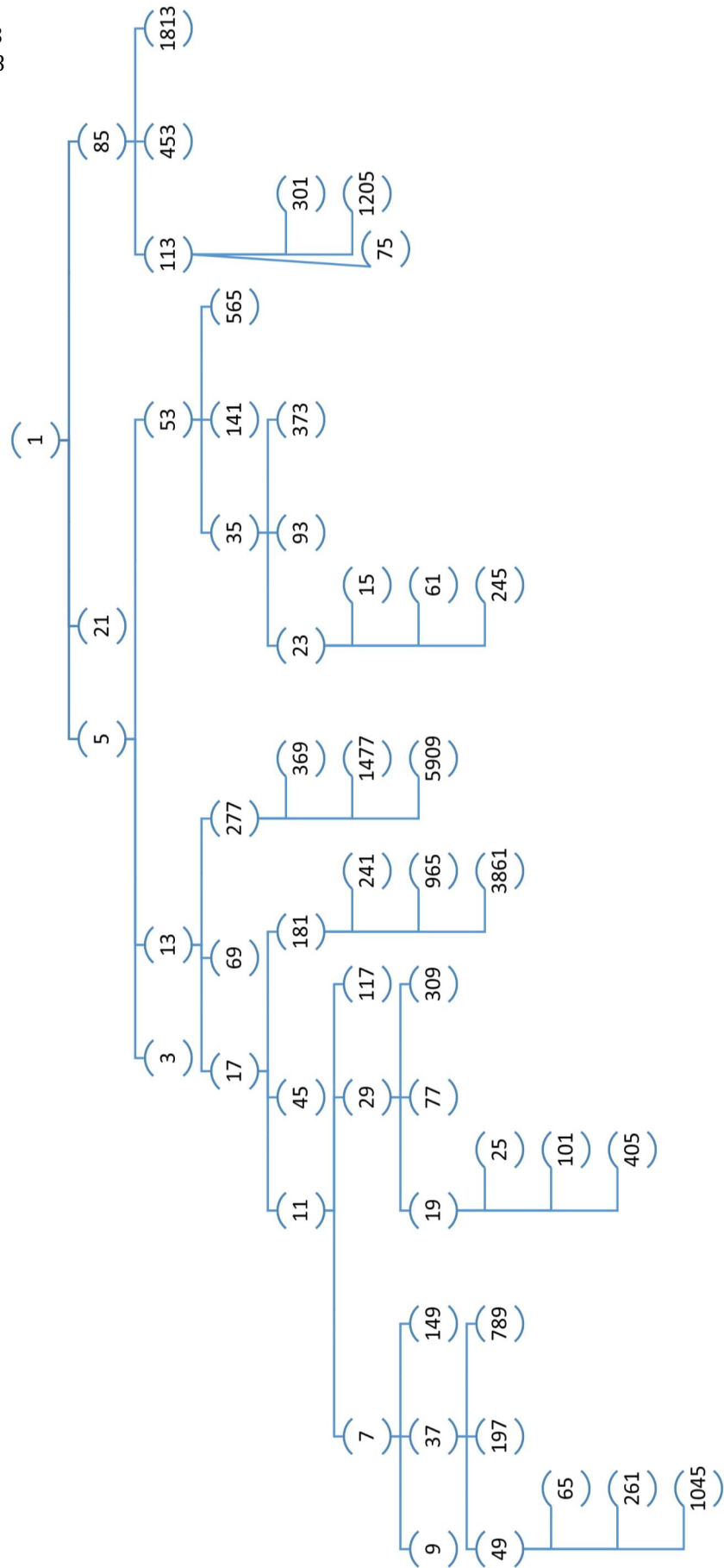


Figure 1: Optimization Tree Illustrating Collatz Convergence Patterns for Odd Numbers

Max =25 ,for $\begin{cases} C = 3 \\ L = 8 \end{cases}$



Future Solution: Demonstrating Convergence Proof through Enhanced Optimization Tree Parameters

In the future development of the proposed optimization technique, emphasis will be placed on validating the convergence proof for all odd numbers within the optimization tree. To achieve this, the optimization tree will be parameterized with two critical factors: the number of children nodes of every node C and the number of layers L .

Enhancement Parameters:

1. C - Number of Children Nodes:

- The parameter C represents the number of children nodes stemming from each node in the optimization tree. A higher C signifies a broader exploration of potential convergence paths for odd numbers.

2. L - Number of Layers:

- The parameter L denotes the number of layers in the optimization tree. Increasing L extends the depth of exploration within the tree, enabling a more comprehensive analysis of convergence patterns.

Objective: Maximizing Sequence Representation (Max):

The key objective is to maximize the representation of Collatz sequences within the optimization tree. The concept of "Max" is introduced, signifying the maximum number in a given sequence. In the context of the Collatz Conjecture, Max represents the highest odd number present in the optimization tree.

Validation Criteria:

- The future solution seeks to establish a direct correlation between the enhanced parameters C and L and the resulting Max within the optimization tree.

- The hypothesis is that by increasing both C and L , a larger Max will be observed. A higher Max indicates a more exhaustive representation of Collatz sequences, supporting the assertion that all odd numbers converge to 1.

Formal Statement:

The future solution posits that as both C and L increase, the resulting Max in the optimization tree will proportionally grow. This phenomenon substantiates the comprehensiveness of the convergence proof, affirming that the Collatz sequences for all odd numbers are effectively and reliably demonstrated within the optimization tree. The validation of this relationship between parameters and Max serves as a pivotal step in establishing the effectiveness and reliability of the proposed optimization technique in proving the Collatz Conjecture for odd numbers.

6. Supplementary Examination:

Observation on Odd Numbers Matrix: Waves and Exclusion of Numbers Written as $3(2n + 1)$:

In the context of the Odd Numbers Matrix, a notable observation emerges regarding the exclusion of numbers expressible as $3(2n + 1)$ from the first column. Specifically, these odd numbers are deliberately absent from the initial column, signifying that within the Collatz Conjecture, no odd number returns to them directly.

However, their deliberate inclusion in the subsequent columns creates a distinctive pattern akin to waves. As the Collatz sequence progresses through iterations, the positions of numbers written as $3(2n + 1)$ within the matrix generate undulating formations. This wave-like pattern, defined by the iterative return of these excluded odd numbers in later columns, introduces a dynamic element to the overall structure of the Odd Numbers Matrix.

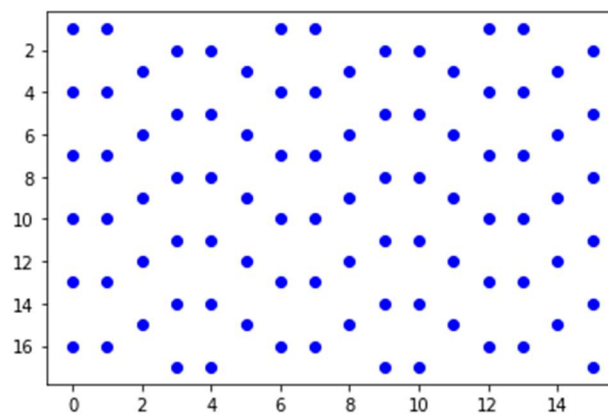


Figure 2: Waves Graph Depicting Undulating Patterns of $3(2n + 1)$